

CENG218 – Labwork 2

The goal of this lab is to understand and implement dynamic arrays in C++ while applying CRUD (Create, Read, Update, Delete) operations. This will help you learn memory management, dynamic resizing, and structured programming using functions.

Your program should have the following functionalities:

- Dynamic Memory Allocation: Use “new” and “delete[]” to allocate and deallocate memory dynamically. Initialize the array with a default capacity of 5 elements.
- CRUD Operations: Implement a menu-driven program where the user can perform:
 - o “Create” – Add a new element to the array (resize if full).
 - o “Read” – Display all elements (mark deleted elements as “[deleted]”).
 - o “Update” – Modify an existing element (prevent updating deleted elements).
 - o “Delete” – Remove an element (mark as deleted instead of shifting)
- Dynamic Resizing: When the array is full, double its capacity and copy existing elements.
- Memory Management: Ensure all dynamically allocated memory is properly released before the program exits.

Your solution should include the following functions:

- void initializeArray() – Allocate and initialize memory (set values to “-1”).
- void menu() – Function to display options.
- void create(int value) – Add an element, resize if necessary.
- void read() – Print all elements (show “[deleted]” for removed ones).
- void update(int index, int newValue) – Modify an element only if it exists.
- void deleteElement(int index) – Set the element to “-1” (instead of shifting).
- void resizeArray() – Double the array size when full.

Ensure the main function calls the appropriate functions based on user input.

Note: Please submit your solution in the following format: StudentID_lab2.cpp

Sample Run:

----- MENU -----

1. Create (Add Element)
2. Read (List Elements)
3. Update Element
4. Delete Element
5. Exit

Enter your choice: 1

Enter the value to add: 10

Element added successfully!

Enter your choice: 1
Enter the value to add: 20
Element added successfully!

Enter your choice: 2
Array Elements: 10 20

Enter your choice: 4
Enter the index to delete (0-1): 1
Element deleted!

Enter your choice: 2
Array Elements: 10 [deleted]

Enter your choice: 5
Exiting program...
Memory freed.